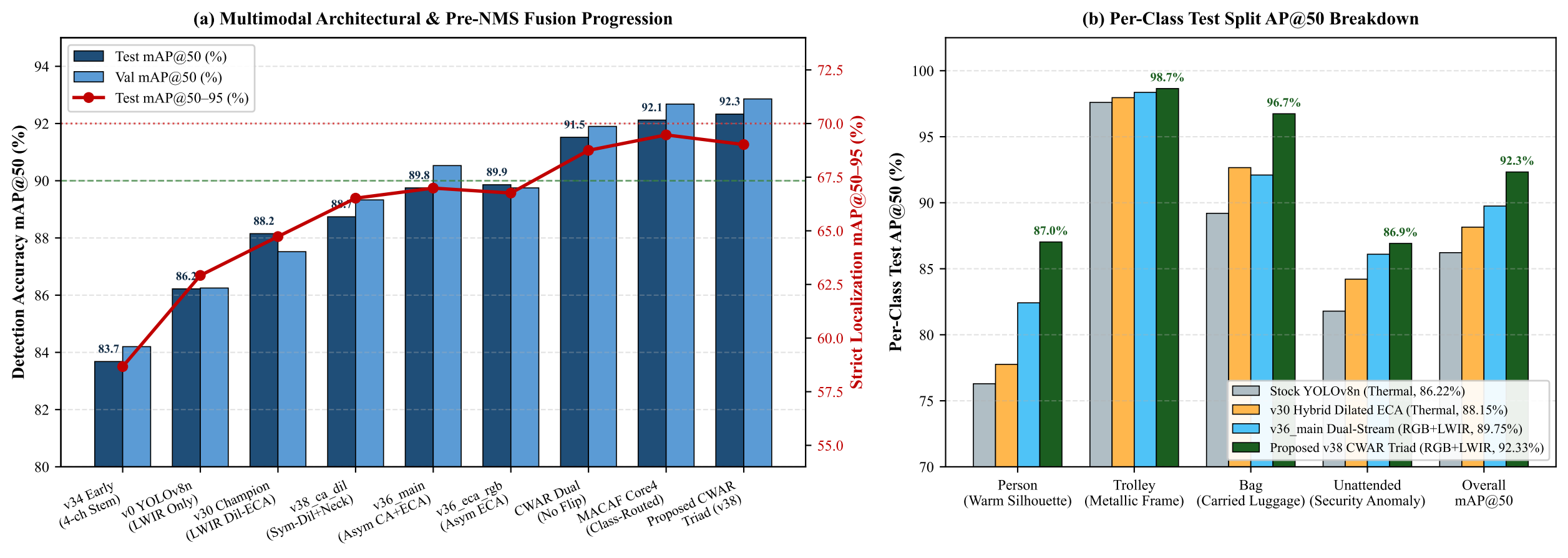

(a) Multimodal Architectural & Pre-NMS Fusion Progression
Test mAP@50 (%)
Val mAP@50 (%)
Test mAP@50–95 (%)
Detection Accuracy mAP@50 (%)
Strict Localization mAP@50–95 (%)
83.7
86.2
88.2
89.7
89.8
89.9
91.5
92.1
92.3
v34 Early (4-ch Stem)
v0 YOLOv8n (LWIR Only)
v30 Champion (LWIR Dil-ECA)
v38_ca_dil (Sym-Dil+Neck)
v36_main (Asym CA+ECA)
v36_eca_rgb (Asym ECA)
CWAR Dual (No Flip)
MACAF Core4 (Class-Routed)
Proposed CWAR Triad (v38)
(b) Per-Class Test Split AP@50 Breakdown
Per-Class Test AP@50 (%)
87.0%
98.7%
96.7%
86.9%
92.3%
Stock YOLOv8n (Thermal, 86.22%)
v30 Hybrid Dilated ECA (Thermal, 88.15%)
v36_main Dual-Stream (RGB+LWIR, 89.75%)
Proposed v38 CWAR Triad (RGB+LWIR, 92.33%)
Person (Warm Silhouette)
Trolley (Metallic Frame)
Bag (Carried Luggage)
Unattended (Security Anomaly)
Overall mAP@50